[INCH-POUND]

A-A-59120
22 September 1997
SUPERSEDING
MIL-G-83984
14 February, 1974

COMMERCIAL ITEM DESCRIPTION

GAUGE, SURFACE PLATE FLATNESS CHECKING

The General Services Administration has authorized the use of this commercial item description(CID) as a replacement for MIL-G-83984 for all federal agencies.

1. SCOPE

1.1 Scope. This Commercial Item Description (CID) covers a gauge for checking surface plate wear, including a master reference gauge calibrator, complete with protector cases and covers.

1.2 Intended Use. The intended use of the surface plate flatness checking gauge covered by this CID is the calibration of plane surfaces and the periodic checking of surface plate wear and condition.

2. SALIENT CHARACTERISTICS

2.1 Design. Each gauge shall be new and of the quality and design currently being offered by manufacturers as an industry accepted standard. The gauge assembly shall include a calibrated straight edge and a dial indicator.

2.1.1 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of component parts.

2.1.2 Material. All parts used in fabrication of the gauge shall be new and of the quality necessary to produce a gauge assembly to meet the requirements described herein.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: WR-ALC/LKJE, 460 Second Street, Ste 221, Robins AFB, GA 31098-1640.

AMSC N/A FSC 5220

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:02Z
Check the source to verify that this is the current version before use.

2.2 Construction. The gauge assembly shall be complete with all necessary components and calibrating equipment so that when assembled and calibrated it will accurately perform the gauging operation for which it is designed. Construction shall be free from any characteristics or defects that will prevent the gauge assembly from passing any of the examinations and tests described herein.

2.2.1 Castings and forgings. All castings and forgings shall be free from defects. In no event shall such processes as welding, peening, plugging, filling with cold solder or metallic pastes be used for reclaiming any part of the gauge assembly or calibrating equipment.

2.2.2 Fastening devices. All screws, pins and bolts shall be installed with means for preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked or otherwise permanently deformed.

2.2.3 Surface finish. All surfaces shall be smooth and free from burrs and sharp edges. Surfaces that are subject to corrosion when exposed to normal use shall be protected against such corrosion in a manner that will ensure compliance with this CID. The protective coating shall not loosen, chip, crack or scale with normal use and age.

2.3 Components.

2.3.1 Dial indicator. The dial indicator shall be designed to meet the requirements of American Gauge design standard, ASME/ANSI B89.1.10M-1987 or an equivalent gauge design standard which will meet or exceed the requirements described herein. Dial graduations on the indicator shall be for not greater than twenty millionths of an inch(0.000020”). Measurement accuracy of the indicator shall be within one graduation of the dial indicator. Total full scale dial range of the indicator shall be at least two thousandths of an inch (minus 0.001 to plus 0.001 minimum). Full scale deflection of the indicator shall not require more than four (4) ounces of force applied to the contact point. The indicator or the indicator mounting shall have minute adjustment provisions for zeroing dial indications (a loading is allowable on the spindle when this adjustment is made). The dial indicator shall have the capability of indicating surface variations of 0.000020 inches.

2.3.1.1 Contact point. Unless otherwise specified, contact points shall be either ruby or a material of equal or better wear resistance and hardness when compared to ruby.

2.3.2 Indicator mounting bar or block. The bar or block shall be designed to provide rigid reliable support for the indicator. An extension handle or other suitable means shall be provided to minimize temperature change and possible erroneous measurement as a result of handling the indicator mounting block. The bar or block shall be provided with a minute adjustment feature for positioning the indicator.

2.3.3 Calibrated straightedge. The straightedge shall be used in conjunction with the block mounted indicator. The straightedge shall be equipped with not less than one fixed support and one adjustable support. The height of the fixed support shall correspond with the height of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:02Z
Check the source to verify that this is the current version before use.

block mounted indicator. The adjustable support shall be provided with a minute precise adjustment feature for positioning both ends of the straightedge in a reference plane that is parallel to the surface being checked.

2.3.3.1 Straightedge sizes and accuracies. The overall guaranteed flatness accuracy shall be within the tolerances specified in table I for the respective straightedge sizes shown. The size of straight edge required shall be as specified (see 6.3). This size is the useful length of the straightedge that is applicable.

TABLE I. Straightedge size and accuracies.

Straight-edge Length (Inches)	Flatness Tolerances (Inches)
24	0.000050
36	0.000070
48	0.000130
60	0.000180

2.4 Nameplate. A corrosion resistant metal nameplate, permanently and legibly marked with the following information, shall be securely attached to each gauge assembly.

Nomenclature
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number
Date of Manufacture
Contract or Order number
Federal Stock Number
US

3. QUALITY ASSURANCE PROVISIONS

3.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4. PACKAGING

4.1 Preservation and Packaging. Preservation, packing, and marking shall be as specified in the contract or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:02Z
Check the source to verify that this is the current version before use.

5. NOTES

5.1 Addresses for obtaining copies of referenced documents.

American Society of Mechanical Engineers(ASME) / American National Standards Institute (ANSI) B89.1.10M-1987 - Dial Indicators (For Linear Measurements)

(Application for copies should be addressed to the American National Standards Institute, 11 W 42nd St., New York, NY 10036 or to the American Society of Mechanical Engineers, 345 E 47th St., New York, NY 10017. Non-government publications are also available in or through libraries, or other informational services.)

5.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this commercial item description.
b. Contact point material, if different (see 2.3.1.1).
c. Straightedge useful surface length required (see 2.3.3.1-and Table 1).

5.3 National Stock Numbers (NSN's). The following is a list of NSN's assigned which meet the requirements of this CID.

5220-01-046-3529
5220-01-033-0124

5.4 Source. The following is a source known to meet the requirements of this CID.

Rahn Granite Surface Plate Co.
75 Holiday Dr.
Englewood, OH 45322-2706
CAGE: 94182

MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AR

Reviewers:
Army - AT

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity
Air Force - 84

Agent Activity:
Air Force - 99

(Project 5220-0234)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:02Z
Check the source to verify that this is the current version before use.